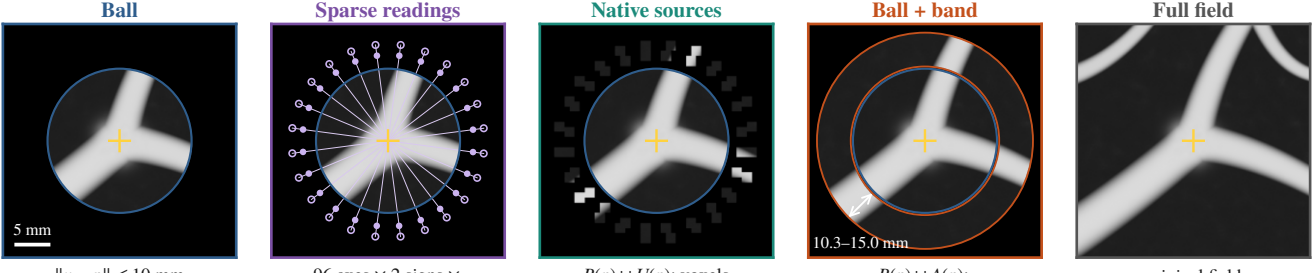
(a) Visibility conditions at one proposal (axial section, to scale)



$\|v - p\| \le 10$ mm

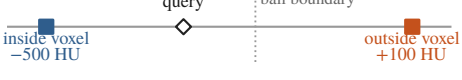
96 axes × 2 signs × 2 radii = 384 values

$B(p) \cup U(p)$: voxels feeding the 384 values

$B(p) \cup A(p)$: complete shell

original field

(b) Hidden voxels become air before resampling (1D example; the query lies 37.5% of the way to the outside voxel)



Interpolate first: $0.625\,(-500) + 0.375\,(+100) = -275$ HU, so outside signal leaks in

Mask first: $0.625\,(-500) + 0.375\,(-1000) = -687.5$ HU, so only fixed air enters